

Project Sentinel

Prospective Prediction of Sepsis-Associated Acute Kidney Injury (SA-AKI)
Technical Report — Full MIMIC-IV Training Run

Headline result: Stage-2 LightGBM @ 24 h achieves **AUROC 0.885** (AUPRC 0.77, calibrated ECE 0.008) on a held-out test set of 4,480 ICU stays / 359,603 patient-hours from full MIMIC-IV v3.1.

Undergraduate capstone research project · African Leadership University
Data: MIMIC-IV v3.1 (credentialed, PhysioNet) · Report generated 2026-07-21

Executive Summary

Serum creatinine is a **lagging** indicator: a patient can lose over half their kidney function before the standard KDIGO criteria flag AKI. Project Sentinel is an end-to-end machine-learning pipeline that predicts Sepsis-Associated AKI **prospectively** — estimating the probability that an ICU patient will meet KDIGO Stage ≥ 1 criteria within the next 6, 12, or 24 hours — so a clinician receives a calibrated, explainable warning *before* the damage shows in the labs.

This report documents the complete pipeline and the results of the first full-scale training run on a 30,000-stay sample of MIMIC-IV v3.1, **before** the planned model-improvement work. It is a reproducible baseline against which future changes are measured.

Key outcomes

- **Task type:** binary classification with a calibrated probability output (not regression).
- **Discrimination:** AUROC 0.885 @ 24 h; 0.934 / 0.902 / 0.885 at the 6 / 12 / 24 h horizons.
- **Calibration:** isotonic-calibrated ECE 0.008 (raw model 0.14) — probabilities mean what they say.
- **Safety:** a split-conformal layer abstains (“INDETERMINATE”) on ~17% of borderline cases instead of guessing.
- **Fairness:** elderly-vs-adult AUROC gap of just 0.02, down from 0.12 on the earlier 100-patient demo.
- **Deployed:** FastAPI + React ward dashboard, one-URL Docker image, ~4.4 ms per alert.

1. Clinical Problem & Motivation

Sepsis-Associated AKI (SA-AKI) is acute kidney injury arising in the context of sepsis. It is common in the ICU and carries high mortality. The diagnostic standard, **KDIGO**, is defined so that by the time it fires, injury has already occurred:

- **Creatinine criterion:** a rise of ≥ 0.3 mg/dL above baseline, OR $\geq 1.5\times$ baseline.
- **Urine-output criterion:** < 0.5 mL/kg/h sustained for ≥ 6 hours (mask-aware).

The clinical value is in **prospective onset prediction** — flagging risk hours before KDIGO would. The eventual target is a resource-limited Rwandan hospital (RMARTH) running the OpenClinic GA EHR; MIMIC-IV is the training surrogate until local data is available.

2. Data & Cohort

MIMIC-IV v3.1 is a credentialed relational ICU database (Beth Israel Deaconess). It is the correct surrogate for this task because — unlike the PhysioNet-2019 Challenge set — it contains **urine output** (icu/outputevents), which is required for the full KDIGO staging above.

Cohort funnel

Stage	ICU stays	Patient-hours
Full MIMIC-IV v3.1 (raw)	94,458	8,275,264
After inclusion (age ≥ 18 , LOS ≥ 8 h) + prevalent-AKI exclusion	92,436	8,242,612
Sampled for training (memory-bounded, seed 42)	30,000	~2.7 M
Held-out test split (chronological)	4,480	359,603

The 30,000-stay sample sits well above the ~10–20k “favorable” tier (events-per-variable heuristic against ~197 features), so subgroup analyses are statistically stable. The test split's positive rate at 24 h is **22.9%** (82,172 positive patient-hours).

3. Pipeline Architecture

Seven sequential stages turn raw multi-table records into a served clinical alert. Each writes a Parquet artifact the next reads, so the data source is chosen once (Notebook 01) and flows through.

#	Stage	Key output
01	Data acquisition — load relational tables, reshape to hourly grid, sepsis onset	hourly_cohort
02	Cohort definition — inclusion criteria, baseline creatinine, prevalent-AKI exclusion	cohort_defined
03	Feature engineering — missingness, rolling windows, creatinine velocity, composites	engineered_features
04	Label engineering — KDIGO labels from RAW data, then chronological split	train/val/test
05	Model training — cascaded LightGBM/XGBoost, Optuna, calibration, conformal fit	models/
06	Evaluation — AUROC/AUPRC/ECE, confusion matrix, DCA, subgroups	reports, figures
07	Explainability — SHAP summaries, waterfalls, clinical-alert JSON	sample_alerts

Three design decisions prevent the silent failures common in clinical ML:

- **Never zero-fill urine output.** 0 mL/kg/h means anuria (severe AKI), not “missing.” Unobserved hours stay NaN with an observation mask.

- **Label from raw measurements, not imputed features.** Imputation (ffill/bfill) would pull a future creatinine backward in time and leak it into the label.
- **Label before the split, and split chronologically by patient.** This mirrors real deployment (train on past patients, predict future ones) and prevents temporal leakage.

4. Feature Engineering

The model sees **197 Stage-2 features** (15 for the Stage-1 screener). Raw hourly values are too sparse and flat to learn from, so the pipeline derives:

- **Missingness indicators** — a lab is ordered *because* a clinician is worried; its absence is signal.
- **Rolling statistics** (mean/min/max/std over 6 h and 12 h) — capture trend, not just the latest value.
- **Creatinine velocity** — rate of rise vs baseline; the single most predictive AKI feature.
- **Composites** — BUN:creatinine ratio, anion-gap proxy, MAP×creatinine-velocity, hours-since-sepsis.

Leakage fix applied this run: the raw `stay_id` identifier was being included as a feature (the model could memorise specific stays). It is now excluded — 198 → 197 features.

5. Model Architecture

A **cascaded two-stage** design mirrors clinical workflow:

- **Stage 1 (screener):** LightGBM on vitals + demographics only — always available, even without a lab. High recall, low latency.
- **Stage 2 (confirmer):** LightGBM + XGBoost on all 197 features (incl. labs), Optuna-tuned — high precision, reduces alarm fatigue.

Separate models are trained per horizon (6/12/24 h). Raw scores pass through **isotonic calibration** (so 0.7 means a real 70% risk), then a **split-conformal (LAC) abstention** layer flags scores inside the band $[1-q_\alpha, q_\alpha]$ as INDETERMINATE. On this run $q_\alpha = 0.630$, giving a band of $[0.370, 0.630]$ and a test-set abstention rate of **17.2%**.

Note: this run used **15 Optuna trials** (a fast validation pass). The full 50-trial tuning is the first planned improvement.

6. Results

All figures below are for the **Stage-2 LightGBM @ 24 h** model on the held-out test set (4,480 stays / 359,603 patient-hours), unless noted.

6.1 Headline metrics

Metric	Value	Interpretation
AUROC	0.885	Strong ranking of at-risk patients; not fake-perfect (no leakage).
AUPRC	0.770	Far above the 22.9% positive base rate.
Brier score	0.128	Overall probabilistic error (lower is better).
ECE (calibrated)	0.008	Near-perfect calibration; raw model 0.14.
Accuracy @0.5	0.821	Misleading under imbalance — see §7.
Precision (PPV) @0.5	0.582	A fired alert is correct ~58% of the time.
Recall / Sensitivity @0.5	0.765	Catches ~77% of AKI onsets.
Specificity @0.5	0.837	Few false alarms among the healthy.
NPV @0.5	0.923	A non-alert is reassuring 92% of the time.
F1 @0.5	0.661	Harmonic mean of precision and recall.

6.2 Confusion matrix (threshold 0.5)

	Predicted NEGATIVE	Predicted POSITIVE
Actual NEGATIVE	TN = 232,212	FP = 45,219
Actual POSITIVE	FN = 19,320	TP = 62,852

The model misses 19,320 AKI onsets at this threshold. Lowering the alert threshold to 0.35 raises recall to 0.855 (misses only 11,934) at the cost of precision 0.474 — the operating point is a clinical choice (§7).

6.3 Discrimination by horizon & cascade stage

Horizon	Stage-1 screener AUROC	Stage-2 AUROC	Stage-2 AUPRC
6 h	0.630	0.934	0.764
12 h	0.629	0.902	0.747
24 h	0.631	0.885	0.770

The vitals-only screener (~0.63) is much weaker than the full model (0.89–0.93) — this is precisely why Stage 2 adds laboratory features. Discrimination rises as the prediction window shortens.

6.4 Subgroup fairness

Subgroup	n patient-hours	AKI events	AUROC	ECE
Adult (<65)	194,088	39,821	0.894	0.133
Elderly (65+)	165,515	42,351	0.873	0.153

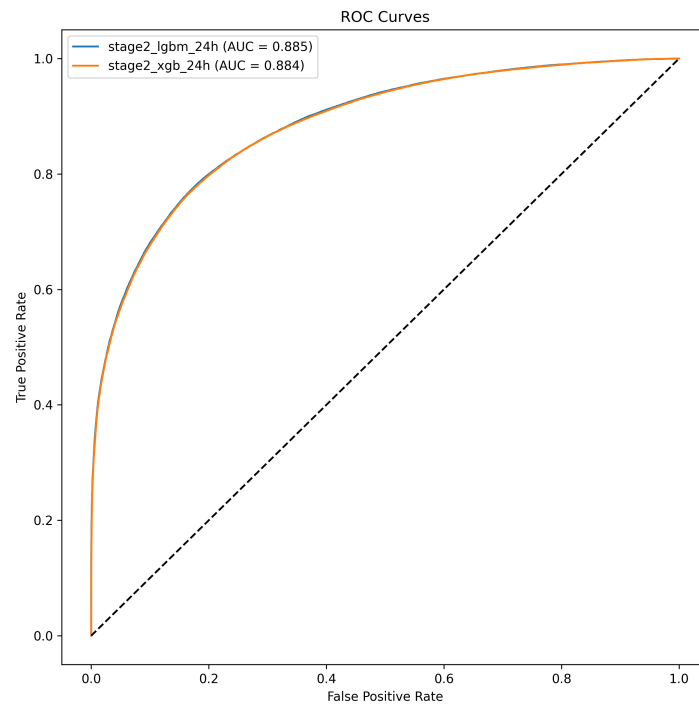
The elderly-vs-adult AUROC gap is 0.021 — down from 0.12 on the 100-patient demo. Scaling the data narrowed the disparity, as the sizing analysis predicted. Elderly calibration remains marginally worse.

6.5 Explainability

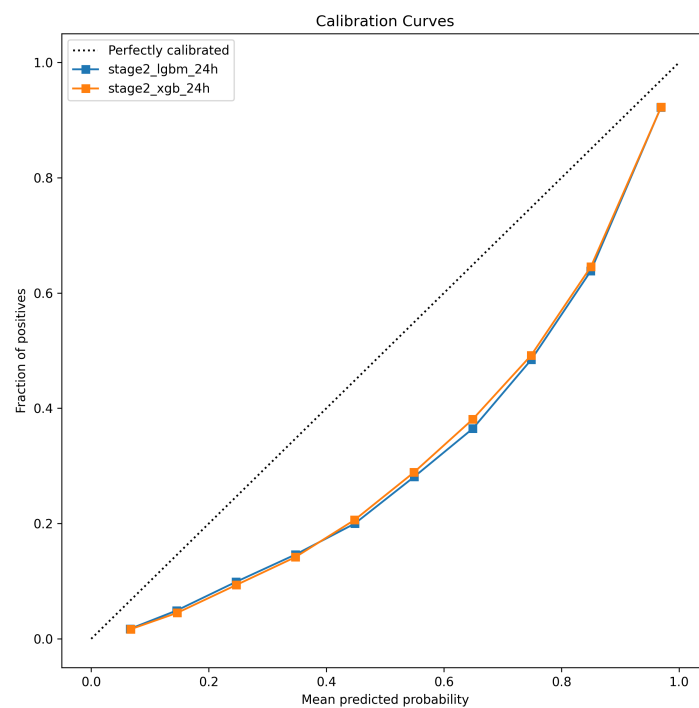
Top SHAP drivers of the 24 h risk score are clinically correct — the model learned physiology, not artifacts: **creatinine-above-baseline, urine rate, 12 h creatinine maximum, creatinine:baseline ratio, baseline creatinine, age**.

6.6 Figures

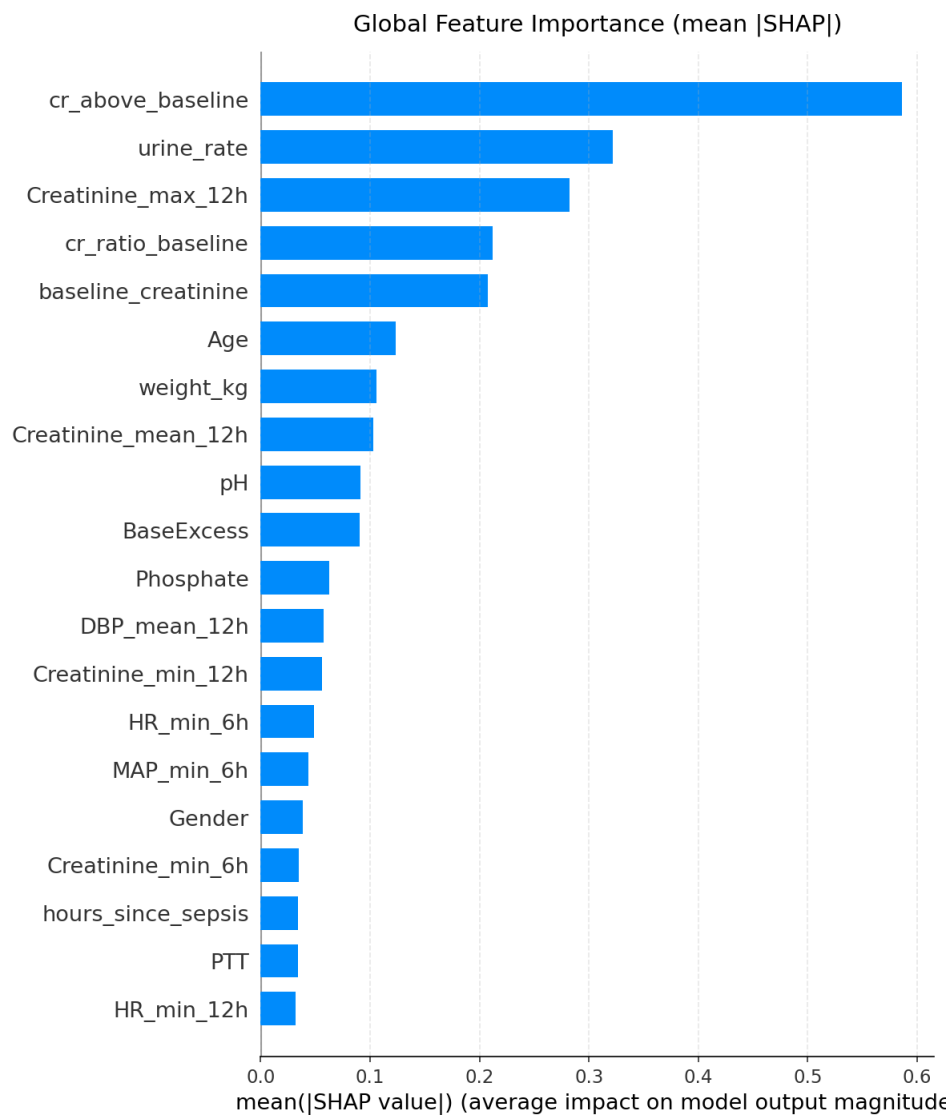
ROC curves (LGBM vs XGB):



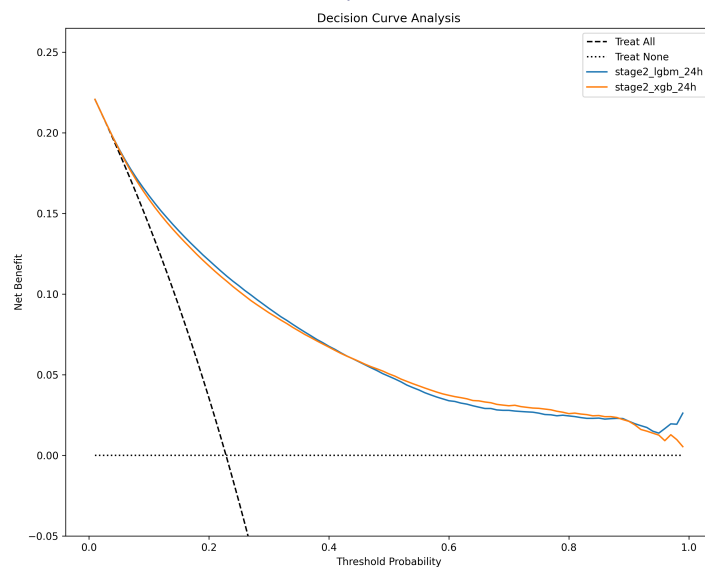
Calibration curves — predicted vs observed frequency:



Global SHAP feature importance (mean |SHAP|):



Decision-curve analysis (net clinical benefit vs treat-all / treat-none):



7. Methodology — Why These Metrics

Classification, not regression. The target is a binary event (AKI within the horizon: yes/no). The model outputs a probability via `predict_proba`, which is thresholded into POSITIVE/NEGATIVE. The continuous score is what enables calibration, DCA, and the conformal band.

Why not lead with accuracy? The data is imbalanced (22.9% positive). A trivial “always predict NO AKI” model scores 77.1% accuracy while being clinically useless. Our 82.1% is barely above that on accuracy alone — yet AUROC 0.885 shows the model is far better. Accuracy hides this; **AUROC, AUPRC, and calibration do not**, which is why they are the primary metrics.

The threshold is a clinical lever, not a fixed truth. For an early-warning system, a missed AKI (false negative) is usually worse than a false alarm, so the operating threshold is lowered to buy recall:

Threshold	Recall (catch AKI)	Precision	Specificity	Missed AKI (FN)
0.50	0.765	0.582	0.837	19,320
0.35 (alert threshold)	0.855	0.474	0.719	11,934

The conformal layer complements this by abstaining on the uncertain middle rather than forcing a call.

8. Deployment

The model is served as a product, not just a notebook: a two-stage Docker image builds a React dashboard and runs a FastAPI inference API from a single URL (Hugging Face Docker Space). The backend loads the Stage-2 24 h model + a SHAP TreeExplainer once at startup and returns a full clinical alert per request.

Property	Value
Endpoints	GET /health, GET /patients, GET /sample, POST /predict
Latency per alert (mean / p50 / p95)	4.41 / 4.29 / 4.91 ms
Throughput	~227 requests/s
Automated tests	17 passing (clinical-logic, API integration, benchmark)
Deploy	one command: <code>deploy/push_to_hf.sh <space-url></code>

9. Known Limitations

- **Single-source (US ICUs).** Trained/tested on MIMIC-IV only; no external validation yet.
- **30k-stay sample**, not all ~90k (memory-bounded); gain from the rest is marginal.
- **Baseline creatinine** uses the in-stay pre-sepsis minimum, not 7–365-day pre-admission outpatient values.
- **Sepsis onset** uses the suspicion-of-infection half of Sepsis-3 only (SOFA ≥ 2 not yet added).
- **No medication or novel-biomarker features** (nephrotoxic drugs, NGAL, TIMP-2-IGFBP7).
- **In-memory demo feed**, not a live OpenClinic GA EHR connection.

10. Improvement Roadmap

Ranked by return on effort. These are the changes to make **after** this baseline is locked.

Lever	Effort	Expected gain
Run the full 50-trial Optuna (this run used 15)	Trivial — rerun NB05	Small but free; the immediate next step.
Richer features: SOFA, nephrotoxic drugs, novel biomarkers (NGAL, TIMP-2-IGFBP7)	Medium	Biggest real gain — strong AKI predictors not yet present.
Better baseline creatinine (7–365-day pre-admission outpatient)	Medium	More accurate labels → cleaner signal.

Sequence models (GRU/LSTM/Transformer on the raw trajectory)	High	Roadmap deep-net phase; may capture temporal patterns trees miss.
Scale 30k → all ~90k stays	Medium	Marginal — already past the favorable sizing tier.
External validation on eICU-CRD / AmsterdamUMCdb	Medium	Higher value than raising AUROC — proves generalisation.

Honest framing: AUROC 0.885 is already in line with published SA-AKI early-warning models (~0.75–0.85). Chasing 0.90+ on the same single-source data risks overfitting to US-ICU quirks; proving the model holds on a *different* hospital is worth more than squeezing MIMIC.

Appendix — Reproducibility & Glossary

Environment: Python 3.12; LightGBM 4.6, XGBoost 3.2, scikit-learn 1.8, SHAP 0.52. Training on Google Colab Pro (High-RAM, CPU). Full MIMIC-IV never committed to git (DUA). Seeds fixed (42) for the sample and splits. Metadata sidecar (model_metadata.json) records feature lists, library versions, git commit, and q■.

Glossary: AUROC — area under the ROC curve (ranking quality). AUPRC — area under precision-recall (imbalance-aware). ECE — expected calibration error. PPV — positive predictive value (= precision). Sensitivity — = recall. NPV — negative predictive value. DCA — decision-curve analysis (net clinical benefit). Conformal q■ — abstention threshold from split-conformal calibration.